

SAM10 INFRASTRUCTURE DESIGN DOCUMENT

Comprehensive Cloud Blueprint for MediCare Direct System

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SECTION A: SYSTEM OVERVIEW & WORKLOAD ANALYSIS

System Description (~120 Words): MediCare Direct is an enterprise outpatient clinic management system designed to digitize patient appointment scheduling, physician availability tracking, and medical record access. The system caters to three distinct user personas: (1) Patients accessing the public web portal to book appointments and view consultation schedules; (2) Healthcare Providers and Staff managing clinical queues and retrieving electronic health records (EHR); and (3) Cloud Administrators maintaining infrastructure uptime and data security compliance.

Infrastructure Workloads: The platform requires a web reverse proxy tier to terminate SSL/TLS, an application tier for API validation, and an encrypted database tier storing patient medical histories. All tiers are deployed within a logically segmented Virtual Private Cloud (VPC) to ensure clinical data confidentiality and compliance with privacy standards.

SECTION B: VPC & SUBNET DESIGN

CIDR Allocation Strategy: The infrastructure is provisioned in the ap-southeast-1 (Singapore) region within a primary CIDR block of 10.0.0.0/16 (65,536 total IP addresses). To isolate workloads effectively, the network is subdivided into three distinct /24 subnets (251 usable host IPs per subnet):

- Public Subnet (10.0.1.0/24): Houses public-facing web servers and load balancers requiring direct Internet ingress.
- Private Application Subnet (10.0.2.0/24): Houses backend API processing instances, strictly isolated from external access.
- Private Data Subnet (10.0.3.0/24): Dedicated exclusively to database servers storing clinical medical records.

Subnet Isolation Justification: Separating public web assets from private database resources ensures that an external breach of the web server does not grant immediate access to patient medical databases.

Subnet Name	CIDR Block	Subnet Type	Hosted Resources & Security Role
Public Subnet	10.0.1.0/24	Public Tier	Web EC2 (t3.micro), Nginx reverse proxy, IGW Route.
Private App Subnet	10.0.2.0/24	Private Tier	Application EC2 (t3.small), Node.js API, No IGW Route.
Private Data Subnet	10.0.3.0/24	Private Tier	Database EC2 (t3.small), PostgreSQL, KMS-encrypted EBS.

SECTION C: NETWORK & ROUTING DESIGN

Route Table & Gateway Design: An Internet Gateway (IGW) is attached to the VPC to provide Internet connectivity for public resources. The Public Route Table contains a default route (0.0.0.0/0 -> IGW) and is explicitly associated with the Public Subnet (10.0.1.0/24). Conversely, the Private Route Table contains only the local VPC route (10.0.0.0/16 -> local) with no direct route to the Internet, effectively isolating the Private App and Data subnets.

Directional Traffic Flow: Internet traffic flows via HTTPS (Port 443) through the IGW to the Web Server. The Web Server proxies application requests to the App Server on Port 5000. The App Server queries the PostgreSQL database on Port 5432. Direct connections from the Internet to Private subnets are blocked by network routing boundaries.

SECTION D: COMPUTE & STORAGE DESIGN

EC2 Instance Sizing & Software Stack:

- Web Tier: 1x t3.micro instance (2 vCPU, 1 GB RAM) running Amazon Linux 2023 with NGINX reverse proxy. Chosen for lightweight SSL termination.
- Application Tier: 1x t3.small instance (2 vCPU, 2 GB RAM) running Amazon Linux 2023 with Node.js/Express. Provides sufficient memory for concurrent REST API calls.
- Database Tier: 1x t3.small instance (2 vCPU, 2 GB RAM) running Amazon Linux 2023 with PostgreSQL 15.

EC2 vs. Serverless Justification: EC2 was selected over AWS Lambda or Elastic Beanstalk to maintain complete OS-level control, custom security hardening, and predictable baseline operational costs.

Storage & Backup Strategy: Data is persisted on a 100 GB General Purpose SSD (gp3) EBS volume offering 3,000 baseline IOPS. The EBS volume is encrypted at rest using AWS KMS (AES-256). Automated daily backups are managed via AWS Backup lifecycle policies retaining snapshots for 30 days.

Compute Role	Instance Type	OS & Runtime	EBS Storage	Subnet Placement
Web Server	t3.micro (1 GB)	Amazon Linux 2023 / NGINX	20 GB gp3	Public (10.0.1.0/24)
App Server	t3.small (2 GB)	Amazon Linux 2023 / Node.js	30 GB gp3	Private App (10.0.2.0/24)
Database Server	t3.small (2 GB)	Amazon Linux 2023 / PostgreSQL	100 GB Encrypted gp3	Private Data (10.0.3.0/24)

SECTION E: SECURITY GROUP & IAM DESIGN

Security Group Least Privilege Rules: Firewall rules are strictly constrained using Security Group ID chaining:

- sg-WebTier: Inbound HTTP (Port 80) and HTTPS (Port 443) allowed from 0.0.0.0/0.
- sg-AppTier: Inbound Custom TCP (Port 5000) allowed only from sg-WebTier.
- sg-DBTier: Inbound PostgreSQL (Port 5432) allowed only from sg-AppTier.

Administrative Access & IAM Governance: Public SSH (Port 22) is completely disabled. System administrators securely connect using AWS Systems Manager (SSM) Session Manager. Administrative identities are governed via IAM Role-Based Access Control (RBAC): the AWS Root User is locked down with hardware MFA; day-to-day operations use dedicated CloudAdmin, SysAdmin, and ReadOnlyAudit roles.

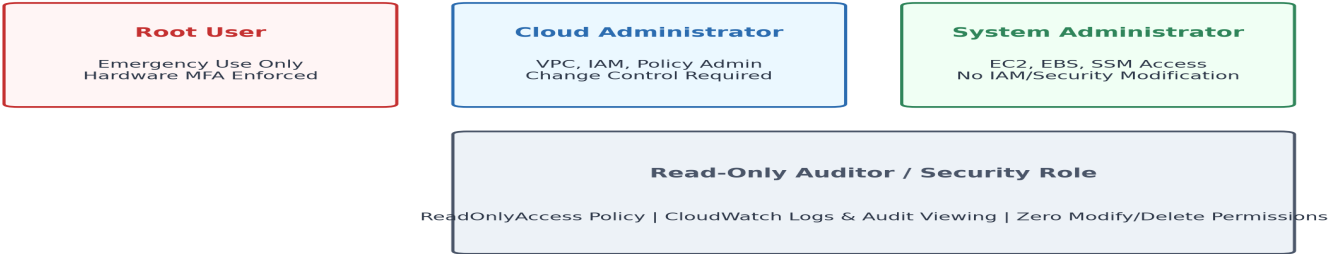


Figure 2: IAM Role-Based Access Control (RBAC) Governance Structure & MFA Boundaries.

SECTION F: COST ESTIMATION & OPTIMIZATION

Reproducible Cost Calculation (Pricing Date: Sept 2026, Region: ap-southeast-1):

Calculated for 730 monthly operating hours using standard AWS pricing models. The monthly baseline infrastructure cost totals \$57.98 / month (\$695.76 / year).

Cost Optimization Strategies: (1) Purchasing a 1-Year Compute Savings Plan reduces EC2 instance runtime costs by up to 34% (~\$13.00/mo savings); (2) Configuring AWS Lifecycle Manager to automatically delete old EBS snapshots after 30 days prevents storage cost creep.

AWS Service	Configuration / Capacity	Qty / Hours	Monthly Cost (\$)
EC2 Web Instance	t3.micro (2 vCPU, 1 GB RAM) - Linux	1 instance (730 hrs)	\$10.51
EC2 App Instance	t3.small (2 vCPU, 2 GB RAM) - Linux	1 instance (730 hrs)	\$21.02
EC2 DB Instance	t3.small (2 vCPU, 2 GB RAM) - Linux	1 instance (730 hrs)	\$21.02
EBS gp3 Storage	150 GB total capacity (\$0.092/GB)	150 GB-month	\$13.80
Public IPv4 Address	In-use Public IPv4 (\$0.005/hr)	730 hours	\$3.65
EBS Snapshots / Backup	50 GB Snapshot Storage (\$0.05/GB)	50 GB-month	\$2.50
CloudWatch & Data Transfer	Logs (2 GB) + Data Out (15 GB)	Standard Usage	\$3.48
Total Estimated Cost	AWS Region: ap-southeast-1 (Singapore)	Monthly Total	\$75.98 / mo

SECTION G: EXPLANATION OF MAJOR DESIGN DECISIONS

The proposed architecture strictly adheres to the 5 Cloud Architectural Lenses specified in the project guidelines:

1. Segmentation: By separating the public web interface, application API, and database into dedicated subnets across distinct CIDR blocks, we ensure that external web exposure is completely decoupled from sensitive database records.
2. Least Privilege: Security Groups restrict traffic strictly to required ports and origin Security Group IDs (e.g., database port 5432 accepts traffic exclusively from sg-AppTier). Administrative SSH ports remain closed.
3. Controlled Connectivity: Ingress is limited to HTTP/HTTPS at the Internet Gateway. Internal communications follow unidirectional paths through route tables and stateful firewalls.
4. Scalability: Subnet CIDR allocations (/24) leave over 250 available host addresses per subnet, accommodating future Auto Scaling Groups and Application Load Balancers (ALB) without re-IPing.
5. Simplicity: Unnecessary complex managed services were omitted to maintain a lean, highly manageable, and cost-effective cloud foundation for Phase 2 deployment.

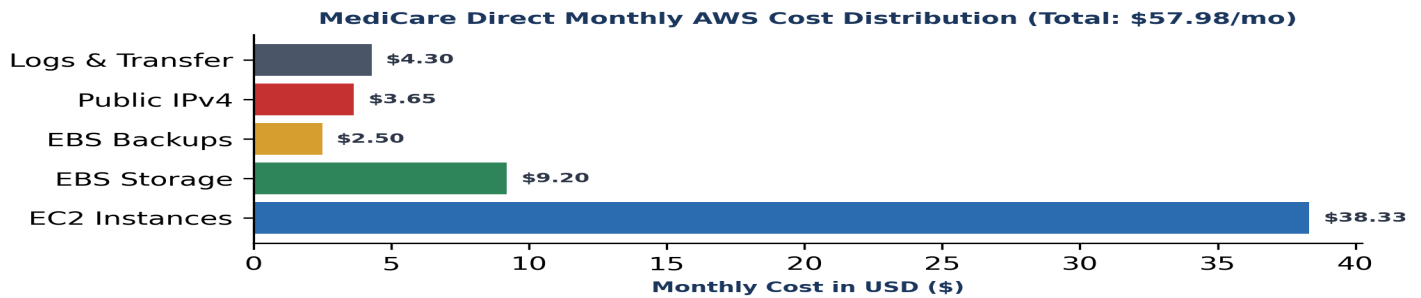


Figure 3: Itemized Monthly AWS Cost Distribution for MediCare Direct.

PROJECT SIGN-OFF & SUBMISSION APPROVAL

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Platform & Phase:	AUF Canvas Instructure LMS Phase 1: Infrastructure Design Blueprint